

Correlation Coefficient

It is a numerical measure (ranging from -1 to 1) that quantifies the strength and direction of a linear relationship between the two variables.

Formula for Correlation Coefficient



Population Correlation Coefficient

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\left(\sum (x_i - \bar{x})^2\right) \left(\sum (y_i - \bar{y})^2\right)}}$$

Where,

$\sigma_x, \sigma_y \rightarrow$ Population Standard Deviation

$\sigma_{xy} \rightarrow$ Population Covariance

$\bar{x}, \bar{y} \rightarrow$ Population Mean

Sample Correlation Coefficient between x and y

$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\left(\sum (x_i - \bar{x})^2\right) \left(\sum (y_i - \bar{y})^2\right)}}$$

Where,

$s_x, s_y \rightarrow$ Sample Standard Deviation

$s_{xy} \rightarrow$ Sample Covariance

$\bar{x}, \bar{y} \rightarrow$ Sample Mean